Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-F Pericardial Disease

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Spread the two brain dumps for a competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-06-C01				
NPLEX-P1-CARD-18-06-C02				
NPLEX-P1-CARD-18-06-C03				
NPLEX-P1-CARD-18-06-C04				
NPLEX-P1-CARD-18-06-C05				
NPLEX-P1-CARD-18-06-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C01-BD1

Pericardial layers and the findings of acute pericarditis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C01 Activity: NPLEX-P1-CARD-18-06-C01-BD1 Date: _____

Without looking at your notes, draw the pericardium in cross section. Label the fibrous sac, the parietal serous layer, the visceral layer, and the space between them, and write the normal volume of fluid in that space and what that fluid is. Write which layers carry pain fibers, which nerve supplies them, and the spinal levels that nerve comes from. Then list the classic findings of acute pericarditis (the pain and what makes it better or worse, where it refers, the friction rub with its components and where you hear it, fever, the blood tests) and write beside each finding the one sentence of mechanism that produces it. Finish by naming the usual cause in developed countries, the classic virus, and the type of exudate that forms.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C01-BD2

Read the tracing back to the inflamed epicardium

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C01 Activity: NPLEX-P1-CARD-18-06-C01-BD2 Date: _____

A 25-year-old man has sharp chest pain that eases when he leans forward. The electrocardiogram shows concave ST elevation in I, II, III, aVF, and V2 to V6 with PR depression in those leads, ST depression and PR elevation in aVR, and no Q waves. Write why the pericardium itself produces none of these changes and which tissue does. Explain why the ST elevation covers more than one coronary territory, why there is no reciprocal ST depression except in aVR, what makes aVR the mirror lead, and what the PR shift tells you about which chamber is involved. Then write the four stages of the electrocardiographic evolution in order. Finish by writing what a raised troponin in this patient would mean and what that condition is called.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-06-C01-CP01

Acute pericarditis against ST elevation myocardial infarction

Fill in every cell from memory, then check your table against the notes and correct anything you missed.

Feature	Acute pericarditis	ST elevation myocardial infarction
Distribution of ST elevation		
Shape of the elevated ST segment		
Reciprocal ST depression		
PR segment		
Q waves		
Timing of T wave inversion relative to the ST segment		
Tissue that generates the ST shift and why the distribution follows from it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-06-C01-Q01

Diffuse ST elevation with PR depression

A 24-year-old man comes to the emergency department because of 2 days of sharp retrosternal chest pain that worsens when he lies flat and eases when he sits forward. He had a diarrheal illness 10 days earlier. Temperature is 37.9 °C (100.2 °F), pulse is 96/min, respirations is 18/min, and blood pressure is 118/74 mm Hg. A scratchy sound is heard at the lower left sternal border. The electrocardiogram shows concave upward ST elevation in leads I, II, III, aVF, and V2 through V6 with PR segment depression in the same leads, ST depression with PR elevation in lead aVR, and no Q waves. Which of the following best explains the distribution of the ST segment changes?

A. Occlusion of a dominant right coronary artery supplying both the inferior and lateral walls

B. Electrical current generated by the inflamed fibrous and parietal pericardial layers

C. Current of injury from inflammation of the epicardial myocardium across the whole cardiac surface

D. Beat to beat change in cardiac axis produced by the heart swinging within pericardial fluid

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C02-BD1

Build the secondary pericarditis table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C02 Activity: NPLEX-P1-CARD-18-06-C02-BD1 Date: _____

On blank paper build a table of the causes of secondary pericarditis with one row for each of these: early post-infarction pericarditis, Dressler syndrome, cardiac surgery, uremia, systemic lupus erythematosus, rheumatoid arthritis, mediastinal radiation, tuberculosis, pyogenic bacteria, and cancer. Use the columns Mechanism of injury and Usual exudate, and fill every cell. Above the table, write the rule that links the kind of injury to the kind of fluid: write what degree and type of injury gives serous fluid, fibrinous fluid, pus, blood, and caseous material. Give the descriptive name used for a heavily fibrinous pericardium. Underneath the table, mark which exudates are the most common and which two are the most likely to organize into constriction.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C02-BD2

Four patients, four exudates

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C02 Activity: NPLEX-P1-CARD-18-06-C02-BD2 Date: _____

Four patients develop pericarditis. Patient 1 is on day 2 after a transmural anterior infarct. Patient 2 is 4 weeks after the same infarct and now has fever, pleuritic pain, a rub, and an effusion. Patient 3 has end stage renal failure and no diffuse ST elevation on the tracing, with bloody fluid drained from the sac. Patient 4 has pneumonia with an empyema and is acutely toxic. For each patient write the mechanism of pericardial injury, the expected exudate, and the one feature of the history that identifies that mechanism. Then explain why patient 1 would not be expected to have pericarditis if the infarct had been subendocardial, why the timing in patient 2 fits an immune mechanism rather than direct injury and what the same picture after cardiac surgery is called, why the tracing in patient 3 can lack diffuse ST elevation and why the fluid is bloody, and which of these four is most likely to heal by dense fibrosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-06-C02-RT01

Rebuild the Dressler syndrome sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C02 Activity: NPLEX-P1-CARD-18-06-C02-RT01 Date: _____

Rebuild from memory the five step sequence that runs from myocardial injury to the clinical picture of Dressler syndrome, one step per line and in order. Beside the step where antigens are released, write where those antigens normally sit and why the immune system has not met them before. Beside the priming step, write how long it takes and which arms of the immune system are involved. Beside the final step, write the findings the patient has and the exudate in the sac. Underneath, write the single feature that separates this process from early post-infarction pericarditis and the name given to the same process after cardiac surgery.

Rebuild from memory: Dressler syndrome as an immune reaction to injury

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C03-BD1

The rigid shell and what it does to filling

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C03 Activity: NPLEX-P1-CARD-18-06-C03-BD1 Date: _____

Write the five step sequence that turns an acute pericardial exudate into a rigid constricting shell, one step per line. Name the major cause worldwide and the three usual causes in developed countries. Then write what happens to diastolic filling inside that shell: describe early diastole, describe where and why filling stops, and name the sound this makes and how it differs from an S3 in timing and pitch. Draw the ventricular pressure tracing of this filling pattern and name it, and draw the jugular venous pulse and mark the descent that is steep. Write what happens to diastolic pressure in each of the four chambers, what happens to systolic function and to stroke volume, and list the signs that follow from the high venous pressure. Finish with the imaging finding that can be seen on a plain film or CT.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C03-BD2

Kussmaul sign and telling the sac from the muscle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C03 Activity: NPLEX-P1-CARD-18-06-C03-BD2 Date: _____

Write what the jugular veins normally do during inspiration and the two steps of mechanism behind that. Then write what they do in constrictive pericarditis, name that sign, and explain it from two things the rigid shell does: what it blocks and what the descending diaphragm still does. Use the same isolation of the heart from intrathoracic pressure to explain why the two ventricles fill reciprocally with breathing, naming what happens to the pulmonary veins and to the septum. Write the two other conditions in which this sign appears. Then a 58-year-old woman has raised venous pressure, ascites, edema, clear lung fields, and a normal ejection fraction. Write the findings you would use to decide between a stiff pericardium and a stiff myocardium, and for each finding write the value or appearance you expect in each condition. Name four diseases that infiltrate or scar the myocardium to produce the second picture.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-06-C03-RT01

Rebuild the path from pericarditis to constriction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C03 Activity: NPLEX-P1-CARD-18-06-C03-RT01 Date: _____

Rebuild the five step sequence that runs from an acute pericardial exudate to a shell that fixes total cardiac volume, one step per line and in order. Beside step one, write which three kinds of exudate are most likely to take this path. Beside the organization step, write which tissue does the organizing. Beside the fusion step, write what happens to the pericardial space and to the normal pericardial fluid. Beside the final step, write what is now fixed and what the myocardium inside is still able to do normally. Underneath, draw the pericardium before and after and write one sentence on what the change does to the pericardial pressure-volume relationship.

Rebuild from memory: From pericarditis to constriction

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C04-BD1

Effusion: the curve, the causes, and the signs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C04 Activity: NPLEX-P1-CARD-18-06-C04-BD1 Date: _____

Write the normal volume of pericardial fluid, then draw the pericardial pressure-volume curve with volume on the horizontal axis and pressure on the vertical axis. Label the reserve volume and the steep limb, and write what is happening in the collagen of the sac along each part of the curve. Draw a second curve for a sac that has been stretched over months and write which way it shifts and why. Then group the causes of effusion by the composition of the fluid: write which causes give a transudate and the Starling change behind each, which cause gives a large protein-rich slowly forming effusion and through what two mechanisms, and which causes give hemorrhagic fluid. Finish by listing the findings of a large effusion, with one line of mechanism beside each, and write which of those findings are not specific and what else causes them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C04-BD2

Two patients, and why volume is the wrong number

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C04 Activity: NPLEX-P1-CARD-18-06-C04-BD2 Date: _____

Patient A has untreated hypothyroidism and 1,200 mL of pericardial fluid that accumulated over four months, with a blood pressure of 124/78 mm Hg. Patient B has a stab wound to the chest and 200 mL of blood in the sac after 20 minutes, with a blood pressure of 82/60 mm Hg and distended neck veins. Place each patient on the pressure-volume curve you drew and explain, for each one, what the sac had time to do and where pericardial pressure sits relative to right atrial pressure, giving the normal value of right atrial pressure. Write which chamber stops filling first and why that chamber rather than another. Then write what the chest film and the electrocardiogram are likely to show in each patient, and why the lung fields are clear in patient A. Finish with the rule this pair of patients teaches about predicting compression.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-06-C04-PR01

Predict the effect of a slowly accumulating effusion

A 52-year-old woman with untreated hypothyroidism has an effusion that has been building for several months and now measures 1,100 mL. For each item below, predict the finding or the direction of change and write the one line of mechanism that explains it.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the shape and position of the pericardial pressure-volume curve as the sac stretches over months		
What happens to pericardial pressure compared with right atrial pressure at this volume		
What happens to diastolic filling and to stroke volume		
What the amplitude of the QRS complexes does in the limb leads		
What the cardiac silhouette looks like on a chest film		
What the lung fields look like and why		
What would change if the same 1,100 mL had collected in one hour instead		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C05-BD1

Tamponade: definition, causes, and the fall in output

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C05 Activity: NPLEX-P1-CARD-18-06-C05-BD1 Date: _____

Write the definition of cardiac tamponade in one sentence that makes clear it is a pressure state and not a volume. List the causes, and for each of the three classic acute causes write the anatomical reason that cause fills the sac: name the chamber penetrating trauma usually enters and why, the usual timing of free wall rupture after a transmural infarct and what has happened in the wall by then, and why a type A aortic dissection can bleed into the sac. Then write the six step sequence from fluid collecting to hypotension with distended neck veins, one step per line, and write which side of the heart is compressed first and why. Write which pressures become equal, name the triad of findings with the mechanism of each part, and name the compensating response and the electrocardiographic finding that may appear.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C05-BD2

Separate fluid under pressure from a fixed shell

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C05 Activity: NPLEX-P1-CARD-18-06-C05-BD2 Date: _____

Write when filling occurs in tamponade and what makes it possible at that moment, then write what the jugular x and y descents look like and explain each from the timing of filling. Now build a comparison of tamponade and constriction on blank paper covering onset, pulsus paradoxus, Kussmaul sign, the y descent, which part of diastole filling is impaired in, whether diastolic pressures equalize, and the extra heart sound, and write one sentence explaining why fluid and a shell differ on the part of diastole they affect. Explain why Kussmaul sign is usually absent in tamponade. Then a patient has a pulsus paradoxus of 24 mmHg. Write how that number is obtained at the bedside with the two pressures you read, write the normal cut-off, explain the finding from ventricular interdependence step by step, and name three conditions other than tamponade that produce it and the mechanism in each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-06-C05-RT01

Rebuild pulsus paradoxus

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C05 Activity: NPLEX-P1-CARD-18-06-C05-RT01 Date: _____

Draw the heart inside a tense fluid-filled sac, then rebuild the seven step sequence from inspiration to a fall in systolic pressure, numbering each step on the drawing and writing it out in words. Mark on the drawing where the septum moves and in which direction, and mark where the pulmonary veins sit and what they do on inspiration. Beside the step where the right ventricle cannot expand, write the property of the sac that prevents it and the name of the principle that links the two ventricles. Beside the last step, write the number of millimetres of mercury that makes the finding abnormal. Underneath, write what happens to total cardiac volume through the respiratory cycle and use that to say in one sentence why right heart gain becomes left heart loss.

Rebuild from memory: Pulsus paradoxus in tamponade

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-06-C05-CA01

Sudden hypotension six days after an anterior infarction

A 71-year-old woman is in the hospital on day 6 after a large anterior ST elevation myocardial infarction that was not reperfused. She suddenly becomes short of breath and loses consciousness. Temperature is 36.8 °C (98.2 °F), pulse is 128/min and weak, respirations are 30/min, and blood pressure is 68/50 mm Hg. The jugular veins are distended to the angle of the jaw, the heart sounds are distant, and the lung fields are clear on auscultation. Systolic pressure falls by 22 mm Hg during quiet inspiration. The monitor shows sinus tachycardia with an organized QRS complex and no palpable pulse. Bedside echocardiography shows an echo-free space around the heart with collapse of the right atrium and right ventricle during diastole.

1. Recognize. Name the mechanical complication that has occurred, the structure that has failed, and the hemodynamic state it has produced, and state why day 6 fits this complication.

2. Explain. Explain why cardiac output has fallen, naming which pressures have equalized, which chambers are compressed first and why, and why electrical activity continues while no pulse is felt.

3. Differentiate. Write how this picture differs from acute left ventricular failure after reinfarction and from constrictive pericarditis, naming the findings in this patient that separate them.

4. Investigate. Write which bedside and echocardiographic findings you would look for to confirm the diagnosis and what each one would show in this condition.

5. Decide. State the property of the pericardial sac that made this volume of blood enough to stop filling, and name the point on the pressure-volume curve the sac has reached.

6. Justify. Justify the inspiratory fall in systolic pressure from ventricular interdependence, and explain why the jugular venous pressure does not rise with inspiration in this patient even though it does in constriction.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-06-C05-Q02

Blunted y descent in a tense sac

A 63-year-old man with metastatic lung carcinoma comes to the emergency department because of 3 days of worsening dyspnea. Temperature is 37.1 °C (98.8 °F), pulse is 122/min, respirations are 26/min, and blood pressure is 88/68 mm Hg with a 19 mm Hg fall in systolic pressure during inspiration. The jugular veins are distended, the heart sounds are distant, and no extra heart sound is heard. The jugular venous waveform shows a preserved x descent and an absent y descent. Right atrial, right ventricular diastolic, and pulmonary capillary wedge pressures are all 22 mm Hg. Which of the following best explains the absent y descent in this patient?

A. Pericardial pressure remains above right atrial pressure after the tricuspid valve opens, so early diastolic filling cannot occur

B. A rigid fibrocalcific pericardium permits very rapid filling in early diastole until the limit of the shell is reached

C. Loss of atrial contraction removes the late diastolic contribution to ventricular filling

D. Impaired myocardial relaxation from amyloid infiltration slows the early diastolic pressure fall in the ventricle

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C06-BD1

How tumor reaches the heart and what it does there

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C06 Activity: NPLEX-P1-CARD-18-06-C06-BD1 Date: _____

Write how common cardiac and pericardial metastases are at autopsy in patients dying of cancer, and how that frequency compares with primary cardiac tumors, naming the usual primary tumor. List the three sites of cardiac involvement in order of frequency. Then build a table of the four routes of spread to the heart with the tumors typical of each, and write one line under the table explaining which routes are explained by proximity and which by wide bloodstream spread. Write which two carcinomas account for the largest number of cases and why, and which tumor involves the heart in a high proportion of patients with widespread disease despite being less common. Finish with the two hematologic malignancies that often infiltrate the heart.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C06-BD2

The effusion that keeps coming back

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C06 Activity: NPLEX-P1-CARD-18-06-C06-BD2 Date: _____

A 66-year-old woman with breast carcinoma has progressive dyspnea, distant heart sounds, and an effusion that reaccumulates two weeks after it is drained. Write the five step sequence from tumor reaching the pericardium to failure of filling, one step per line. Write the two mechanisms by which pericardial deposits produce fluid, name the kind of exudate that results, and write the test on the fluid that confirms tumor as the cause. Explain why a rapidly growing or repeatedly reaccumulating effusion points to tumor, and write why malignancy is a leading cause of nontraumatic tamponade. Then write the two other pericardial problems a patient with cancer can have instead of tumor in the sac, naming the treatment-related one and what tumor encasing the heart can imitate. Finish with what myocardial and intracavitary deposits do.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____